FREELANCE · REMOTE · AWS CERTIFIED · BOOKING NEW PROJECTS

CHINTALA MAHINDRA

Full-Stack, Mobile & Cloud Engineer

From first website to millions of requests.

Websites · Mobile apps · System design · Spring Boot · React · AWS · Docker · Kubernetes · CI/CD

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PROFILE

Full-stack, mobile and cloud engineer who designs, builds and scales products end to end: **React, Next.js and React Native** apps, **Java 21 / Spring Boot 3** services, and **AWS** infrastructure running on **Docker and Kubernetes** with automated CI/CD. I start from the system design (caching, queues, idempotency, isolation, failure modes) and write the decisions down before the code, so clients know what they are getting, why, and what it will cost to run.

WHAT I DELIVER

- **Websites & mobile apps** for small businesses and personal brands: fast, mobile-first sites, React Native apps for iOS and Android, and low-cost care plans.
- **Startup MVPs** for busy founders, taken from idea to production on AWS and built to grow rather than be rewritten.
- **Scale, cloud & CI/CD**: load-test-driven performance fixes, EKS via Terraform and Helm, GitOps with Argo CD and canary releases.
- **AI features**: permission-aware RAG and document Q&A with citations, cost controls and evals.

EXPERIENCE

Tata Consultancy Services

AUG 2024 – PRESENT

Software Engineer, Java Backend · Hyderabad · Enterprise HRMS & finance microservices

- Build and enhance **Spring Boot 3 microservices on Java 21** for payroll, workforce and finance workflows, with REST APIs and Spring Data JPA over PostgreSQL and SQL Server.
- Secured employee, HR and admin endpoints with **Spring Security, JWT and role-based access control**.
- Added **asynchronous event processing** and **Redis caching** for high-frequency payroll and attendance lookups.
- Integrated **Spring AI** into existing services for document and policy Q&A, with access control applied to retrieved context.
- Production debugging and API / database issue resolution across backend services in Agile delivery.

SEE IT LIVE

MAIN-HUB-MU.VERCEL.APP

SYSTEMS LAB
Interactive scaling model: traffic, autoscaling, caching, queues and replicas, with the bottleneck explained.

6 ARCHITECTURES
Clickable diagrams with live request traces, scaling stages, decisions and failure modes.

SHOWCASE
Sample websites and matching mobile apps for small businesses, with clear starting prices.

EXPERTISE

SYSTEM DESIGN
Scalability, caching, queues, idempotency, multi-tenancy, event-driven design, consistency, capacity planning

BACK END
Java 21, Spring Boot 3, Spring Security (JWT, OAuth2, RBAC), REST, WebSockets, microservices

WEB & MOBILE
React, Next.js, React Native (Expo), TypeScript, Tailwind CSS, SEO

DATA
PostgreSQL / Aurora, Redis, DynamoDB, pgvector, SQL Server

AWS
EKS, EC2, Lambda, RDS, S3, CloudFront, SQS, SNS, IAM, Secrets Manager, CloudWatch

PLATFORM
Docker, Kubernetes, Helm, Terraform, Argo CD, GitHub Actions, OpenTelemetry

AI INTEGRATION
Spring AI, RAG, embeddings, vector search, LLM APIs, structured outputs

HOW I WORK

1 · Discovery call
Product, users, pain, deadline.

2 · Design doc & fixed quote
Diagrams, decisions, trade-offs.

3 · Weekly sprints
Working demo on staging every week.

4 · Launch
Load-tested, monitored, rollback rehearsed.

5 · Handover
Runbooks; you own the repo and AWS account.

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SYSTEM DESIGN CASE STUDIES

INTERACTIVE ON THE PORTFOLIO

LEDGERLINE

PAYMENTS & LEDGER

Double-entry, append-only journal; idempotent transfers; transactional outbox; daily reconciliation.

ROUTELIVE

REAL-TIME DISPATCH

Partitioned location stream; Redis GEO dispatch; WebSocket fan-out; PostGIS history.

DOCMIND

PERMISSION-AWARE RAG

Permission filter inside the vector query; cite-or-decline answers; evals gate prompts in CI.

OPEN SOURCE & PROJECTS

Project 1 · TenantForge — Multi-tenant B2B SaaS platform

2026

OPEN SOURCE

Java 21 · Spring Boot 3 · Next.js · PostgreSQL / Aurora · Redis · SQS · S3 · Keycloak · Stripe · EKS · KEDA · Terraform · Argo CD · github.com/ARGOD2213/tenantforge

- Tenant isolation enforced by **Postgres Row-Level Security** (FORCE RLS, non-owner role); **pool-then-silo** tenancy routed by a tenant catalog, migrations per database.
- **Per-tenant rate limits** in Redis (429 + Retry-After); Keycloak SSO with tenant/role claims and per-tenant OIDC/SAML; **idempotent Stripe webhooks**.
- **SQS exports on KEDA-scaled workers**, S3 per-tenant prefixes with KMS; Terraform for EKS and Aurora; 34 Testcontainers tests in CI.

Project 2 · SwiftCart Flash — Flash-sale commerce platform

2026

OPEN SOURCE

Java 21 · Spring Boot 3 · Next.js · Redis + Lua · Aurora PostgreSQL · SQS FIFO · Razorpay / Stripe · EKS · KEDA · CloudFront + WAF · Terraform · github.com/ARGOD2213/swiftcart-flash

- **Atomic Redis/Lua inventory** with timed holds and per-customer and per-phone limits; a **waiting room** with signed entry tokens; stock rebuilt from the order log if Redis is lost.
- **Charge exactly once**: idempotent checkout, verified, de-duplicated Razorpay/Stripe webhooks, late-payment rescue or refund, a reconciler, outbox to an SQS FIFO worker.
- 99 Testcontainers tests; CI runs a **k6 drop (1,500 shoppers, 300 pairs)** and audits it: no oversell, no double charge.

ARGUS — Event-driven industrial IoT monitoring platform

8 SPRINTS

OPEN SOURCE

Java 17 · Spring Boot 3.2 · AWS Lambda · SQS FIFO · SNS · DynamoDB · S3 · Athena · Python · github.com/ARGOD2213/ARGUS

- Always-on **Lambda rule engine** fed by SQS FIFO; CRITICAL / WARNING alerts through SNS, stored in DynamoDB with evidence.
- **Permit-to-Work** state machine, evidence export, shift handover, nine dashboards; a local **predictive ensemble** advises only.
- Architecture locked by **ADRs**: no write path to plant equipment, Athena + S3 for history, edge agent with offline rules; no ECS, ALB or NAT to keep cost lean.

PolicyDocs RAG Service

2025 – PRESENT

Spring Boot 3 · Spring AI · Java 21 · PostgreSQL + pgvector · Redis · Docker · AWS

- Ingestion, chunking, embeddings and **RAG Q&A** over policy documents with structured outputs; **pgvector** retrieval.
- JWT and RBAC constrain retrieval; grounded answers only; Docker on AWS EC2.

Digital Library Platform

2025 – PRESENT

Spring Boot 3 · Java 21 · React · PostgreSQL · Redis · S3 · SES/SNS · Docker · AWS

- REST APIs with JPA and **Flyway**; role-based workflows for readers, vendors and admins; JWT, **Redis caching**, S3, SES/SNS; Docker on EC2 via **GitHub Actions**.

STANDARDS

Infrastructure as code — Terraform for every AWS resource; separate staging and production.

Delivery — CI on every PR, image and dependency scanning, GitOps with Argo CD, canary or blue-green releases.

Observability — OpenTelemetry traces, Grafana or CloudWatch dashboards, alerts on SLOs.

Security — least-privilege IAM roles, Secrets Manager, OWASP checks, encryption at rest and in transit.

Quality — unit and Testcontainers integration tests, k6 load test before launch, OpenAPI docs.

Ownership — your GitHub and AWS accounts, ADRs, runbooks for every alert.

ENGAGEMENTS

Fixed scope

Clear deliverables, fixed quote, paid by milestone.

Sprint based

Two-week sprints, demo at the end of each.

Retainer

Reserved monthly hours as a fractional backend engineer.

CERTIFICATIONS

AWS Certified Cloud Practitioner

CLF-C02 · 2024

Spring Boot 3 & Spring Framework 6

Udemy (Chad Darby) · 2025

REST API Design, Development & Management

Udemy · 2024

EDUCATION

B.Tech, Electronics & Communication Engineering

VIT Vellore · 2020 – 2024